

CHAPTER 12

12.1) HELIUM AS A VAN DER WAALS GAS

(a) We use the approximation (4) for the enthalpy of the van der Waals gas. If we set $Nb = V_\ell/2$ and $2a/b = \tau_{inv}$, this becomes

$$H = \frac{5}{2} N\tau + (NV_\ell/2V)(\tau - \tau_{inv}) \quad .$$

Approximating V by $V \cong N\tau/p$, from the ideal gas law:

$$H \cong \frac{5}{2} N\tau - (pV_\ell/2)(\tau_{inv}/\tau - 1) \quad .$$

This result could also have been obtained directly from (10.76). With it, the numerator in (7) becomes

$$H_{out} - H_{in} = (V_\ell/2)(p_{in} - p_{out})(\tau_{inv}/\tau - 1) \quad . \quad (S1)$$

The denominator in (7) may be written

$$H_{out} - H_{liq} = H(\tau_{in}, p_{out}) - H(\tau_{liq}, p_{out}) + \Delta H \quad ,$$

where the first two terms on the right are the entropies of the gas at the upper heat exchanger temperature and at the condensation temperature, both taken at the output pressure. If the evaporated He, because of its low pressure, is treated as an ideal gas, the ideal gas value $H = 5N\tau/2$ may be used, which leads to (10). Combining (S1) with (10) leads to

$$\lambda \cong \frac{(V_\ell/2)(p_{in} - p_{out})(\tau_{inv}/\tau_{in} - 1)}{\Delta H + \frac{5}{2} N(\tau_{in} - \tau_{liq})} \quad . \quad (S2)$$

(b) Note that (S2) is linear in the input pressure, in clear disagreement with the actual behavior of He as shown in Fig. 12.4, except possibly at uninterestingly low pres-

tures. If we use $p_{in} = 10$ atm, $p_{out} = 1$ atm, $T_{in} = 15$ K and take all remaining values from Table 12.1, we obtain for the numerator in (S2), 35 J mol^{-1} , for the denominator 307 J mol^{-1} , yielding a predicted liquefaction coefficient of $\cong 0.11$, close to what Fig. 12.4 shows.

The bending-over of the curves of Fig. 12.4 can be reproduced qualitatively by not working with the approximation (4) for H, but by using the exact enthalpy of the van der Waals gas,

$$H = \frac{3}{2} N\tau - \frac{2Na^2}{V} + \frac{N\tau}{1 - Nb/V} \quad ,$$

easily obtained from Eqs. (10.74) of the text and (10.1.S2) of this guide. However, the agreement cannot be made quantitative. The van der Waals model is just not accurate enough. This is also seen by comparing the ratio of critical to inversion temperature, for which the van der Waals model predicts, according to (5), $\tau_c/\tau_{inv} = 27/4 = 6.75$, while the actual ratio for He is $51/5.25 \cong 9.7$.

12.2) IDEAL CARNOT LIQUEFIER

(a) The reversible work dW required to cool the ideal gas from T to $T-dT$, with a Carnot refrigerator operating between the lower temperature T and the upper temperature T_0 , follows from (8.12), (8.13), and (8.38d):

$$dW = \frac{dQ}{\gamma_C} = \frac{C_p}{\gamma_C} dT = \frac{5}{2} R(T_0/T - 1)dT \quad . \quad (S1)$$

Here $\gamma_C = T/(T_0 - T)$ is the Carnot coefficient of refrigerator performance, and $C_p = 5N_A k_B/2 = 5R/2$ is the heat capacity at constant pressure of one mole of monatomic ideal gas in conventional units. If (S1) is integrated from T_b to T_0 one finds the gas precooling work

$$W_1 = \frac{5}{2} R[T_0 \log(T_0/T_b) - (T_0 - T_b)] \quad .$$

To liquefy the cold gas at fixed temperature T_b , the latent heat ΔH must be removed, which requires the additional reversible work

$$W_2 = \Delta H/\gamma_C = \Delta H (T_0 - T_b)/T_b \quad .$$

Combining W_1 and W_2 yields (11).

(b) Using $T_0 = 300\text{K}$, and the values in Table 12.1, one finds $W_1 = 20.5 \text{ kJ/mol}$ and $W_2 = 5.8 \text{ kJ/mol}$, whence $W = 26.3 \text{ kJ/mol}$. There are $1000/32.0 = 31.25$ moles in 1 liter and 3600 seconds in an hour, yielding finally $W = 26.3 \times 31.25 / 3600 \text{ kWh/liter} = 0.228 \text{ kWh/liter}$ for an ideal Carnot He liquefier. This is less than 1/20 of the consumption of actual He liquefiers. The principal reason for the very much higher consumption of actual liquefiers is that -- at this time: 1979 -- they utilize the Joule-Thomson effect for the final cooling step. This is a highly irreversible effect that generates large amounts of new entropy within the liquefier, the removal of which requires a large amount of extra work.

12.3) CLAUDE CYCLE HELIUM LIQUEFIER

Warning. Parts (b) and (c) of this problem are best suited for a classroom discussion extending the coverage of the main text. If assigned as homework, it is advisable to forewarn the students that a naive execution of the calculation leads to unrealistic numbers which in practice force a modification of the simple Claude cycle. The students may be asked to suggest modifications that overcome these difficulties.

(a) From Fig. 12.4, at $T_{in} = 15 \text{ K}$ and $p_{in} = 30 \text{ atm}$ we have $\lambda \cong 0.18$. For a gas flow of 1 mol/s and a molar volume of 32 cm^3 (see Table 12.1), the volume liquefaction rate is then

$$\begin{aligned}\dot{V} &= (0.18 \text{ mol/s})(0.032 \text{ liter/mol})(3600 \text{ s/hr}) \\ &= 20.7 \text{ liter/hr} .\end{aligned}$$

The open-cycle cooling load that can be supported by this liquefaction rate, if the liquid He is withdrawn from the liquefier, is given by

$$\dot{Q} = \dot{V}\Delta H/V_\ell = 14.72 \text{ watt} ,$$

with $\Delta H/V_\ell = 0.71 \text{ watt hr/liter}$ from Table 12.1. If the liquefier is used as a closed-cycle refrigerator, the cooling load from Fig. 12.4 is about 58 watt, roughly 4-times the open-cycle cooling load!

(b) The total enthalpy flowing into the liquefier per second is $H(\text{gas in}) = H(T_c, p_c)$, where $H(T, p)$ designates the enthalpy per mole of gaseous He. The enthalpy outflow consists of three parts:

$$H(\text{work out}) = W_e ,$$

$$H(\text{gas out}) = (1-\lambda)H(T_c, p_{\text{out}}) ,$$

$$H(\text{liquid out}) = \lambda H_{\text{liq}} = H(T_{\text{in}}, p_{\text{in}}) - (1-\lambda)H(T_{\text{in}}, p_{\text{out}}) .$$

In the last line, H_{liq} denotes the enthalpy per mole of liquid He, and the second equality follows from (6). For enthalpy inflow and outflow to balance, the expansion work must satisfy

$$\begin{aligned}W_e &= H(T_c, p_c) - H(T_{\text{in}}, p_{\text{in}}) \\ &+ (1-\lambda)[H(T_c, p_{\text{out}}) - H(T_{\text{in}}, p_{\text{out}})] .\end{aligned}\tag{S1}$$

If we approximate the four gas enthalpies by their ideal gas values, $H(T, p) = 5RT/2$, we obtain

$$W_e \cong (5\lambda R/2)(T_c - T_{\text{in}}) .\tag{S2}$$

The work done in isentropic expansion equals the difference in energy of the gas before and after expansion. If the gas enters the expansion engine at $T = T_e$ and leaves

it at the input temperature of the Linde heat exchanger, $T = T_{in}$, the work per mole is

$$W_e = (3R/2)(T_e - T_{in}) ,$$

where we treated the gas as ideal. If we equate this to (S2):

$$T_e = T_{in} + (5\lambda/3)(T_c - T_{in}) . \quad (S3)$$

For our numbers ($T_c = 50^\circ\text{C} = 323 \text{ K}$, $T_{in} = 15 \text{ K}$): $T_e \cong 107.4 \text{ K}$, or $T_e/T_{in} \cong 7.16$. This is an unrealistically high temperature ratio for a single-stage expansion engine. But the pressures to which it leads are even worse: According to (6.64) the pressure ratio during the isentropic expansion must then satisfy

$$p_e/p_{in} = (T_e/T_{in})^{5/2} \cong 137.18 ,$$

corresponding to $p_e \cong 1.37.18 \times 30 \text{ atm} = 4115 \text{ atm}$. Such high pressures are completely unrealistic and no actual He liquefier employs them. Instead, more complicated variations of the Claude cycle are employed, in which only a small fraction of the gas flow through the liquefier is passed through the Linde stage, thus reducing the work that must be extracted per mole of total gas flow, and with it the need for unrealistically high pressures. We ignore these complications here (see Comments below).

(c) The isothermal work required to compress 1 mole of ideal gas from $p = p_{out}$ to $p = p_c$ at $T = T_c$ is readily obtained from (8.20):

$$W_c = N_A T_c \log(V_{out}/V_c) = RT_c \log(p_c/p_{out}) .$$

With our numbers ($T_c = 50^\circ\text{C}$, $p_{out} = 1 \text{ atm}$, including the unrealistically high derived value of p_c): $W_c \cong 22.4 \text{ kJ/mol}$. At a flow rate of 1 mol/s the compressor power is then $\dot{W}_c = 22.4 \text{ kW}$. This number assumes that the liquified gas is made up by fresh He gas supplied at $p = p_{out}$.

With the two values for the cooling load derived in part (a), $\dot{Q} = 14.72$ watt and 58 watt, we obtain for the coefficient of refrigerator performance, $\gamma = \dot{Q}/\dot{W}_C$, the two values $\gamma = 0.66 \times 10^{-3}$ and $\gamma = 2.6 \times 10^{-3}$. The Carnot coefficient of performance is, by comparison, $\gamma_C = T_{liq}/(T_C - T_{liq}) = 1.31 \times 10^{-2}$, about 5-times the estimated value for a closed-cycle Claude refrigerator, and 20-times the value for an open-cycle Claude liquefier.

The comparison with actual liquefiers is easiest by comparing compressor work per volume of liquid, because this quantity is readily obtainable from the published engineering specifications of actual liquefiers. Typical values are between 5 and 10 kWh/liter. From our estimate of $\dot{W}_C$ and from the estimate $\dot{V} = 20.7$ liter/hr of part (a), we would have expected $\dot{W}_C/\dot{V} \cong 1.08$ kWh/liter. Thus we see that our estimates are still optimistic by a factor 5 to 10, due to various engineering inefficiencies contained in actual liquefiers.

Comments. The problem illustrates the performance penalty relative to a (non-existing) reversible He liquefier, that is an inevitable consequence of the use of the highly irreversible Joule-Thomson expansion for the final liquefaction stage. It shows the advantages to be gained if it should in the future be possible to design more nearly reversibly operating expansion engines (presumably turbines) that can work satisfactorily down to the actual liquefaction temperature. The problem also shows that the Claude cycle in its simplest possible form requires unrealistically high pressures, and that it must be modified for He liquefier operation.

It is instructive to pursue this latter point further by assuming that for every mole of gas passing through the Linde stage there are m moles of gas passed through the expansion engine, and that the excess $m-1$ moles leave the

liquefier at the input pressure p_{in} of the Linde stage, rather than the atmospheric pressure p_{out} . There are then two return gas streams at different pressure, but both streams are assumed to flow through efficient heat exchangers, so both leave at the same temperature T_c at which the compressed gas enters the liquefier. For He near or above room temperature, the pressure dependence of the enthalpy is negligible, hence the existence of two gas streams with different output pressures is irrelevant for the enthalpy balance, and this modified Claude cycle is then equivalent to the original Claude cycle, but with the reduced liquefaction coefficient $\lambda' = \lambda/m$. If we assume $m = 10$, we now obtain $\lambda' = 0.018$, that is, less than 2% of the gas entering the liquefier is liquefied, greatly increasing the physical size of compressors, heat exchangers, expansion engines, etc., for a desired liquefaction rate. With this value one easily obtains the following other numbers: $T_e \cong 24.2K$; $T_e/T_{in} \cong 1.62$; $p_e/p_{in} \cong 3.32$; $p_e \cong 100$ atm.

The 10-fold increased gas flow requires a larger compression work, but not proportionately so, because of the greatly reduced pressure differential:

$$\begin{aligned} W_c &= RT_c [9 \log(p_e/p_{in}) + \log(p_e/p_{out})] \\ &= RT_c [10 \log(p_e/p_{in}) + \log(p_{in}/p_{out})] \\ &\cong 41.4 \text{ kJ/mol} \quad , \end{aligned}$$

not quite twice the value calculated earlier.

Actual He liquefiers are more complex than our simple scheme, containing at least two expansion engines, and more complex heat exchanger arrangements than suggested by Fig. 12.1, but they do not invoke additional physical principles.

12.4) EVAPORATION COOLING LIMIT

If $\dot{Q}$ designates the cooling load, then the rate of evaporation, in mols per unit time, is $r = \dot{Q}/\Delta H$, where $\Delta H = 82$ J/mol is the molar heat of vaporization (see Table 12.1). The evaporated gas, if warmed up to room temperature, occupies the volume

$$\begin{aligned} V &= (24 \text{ liter/mol}) \times (760 \text{ torr})/p \\ &= (18,240 \text{ torr liter/mol})/p \end{aligned}$$

where p is the gas pressure. The rate at which gas is removed by a pump with speed S (rated at room temperature) is

$$r = S/V = pS/(18,240 \text{ torr liter/mol}) \quad .$$

Equating the two values of r , inserting the value of ΔH , and solving for p :

$$\begin{aligned} p &= \frac{18,240 \text{ torr liter/mol}}{\Delta H} \frac{\dot{Q}}{S} = \\ &= (222 \text{ torr liter/J}) \times \dot{Q}/S \end{aligned}$$

for $\dot{Q} = 10^{-1}w$, $S = 10^2$ liter/s: $p = 0.222$ torr. According to Table 12.2, this is the equilibrium vapor pressure of He at a temperature somewhere between 0.98 K and 1.27 K, an interval 0.29 K wide. If we simply interpolate logarithmically in this range, we obtain the estimate

$$T = 0.98 \text{ K} + 0.29 \text{ K} \times \log_{10} 2.22 \cong 1.1 \text{ K} \quad .$$

For $\dot{Q} = 10^{-3}w$, $S = 10^3$ liter/s, the same procedure leads to $p \cong 2.22 \times 10^{-4}$ torr, or

$$T \cong 0.56 \text{ K} + 0.1 \text{ K} \times \log_{10} 2.22 \cong 0.59 \text{ K} \quad .$$

We note that, even though the ratio $\dot{Q}/S$ is a factor 10^{-3} lower in the second case, the achievable temperature is less than a factor 2 lower. This insensitivity of the achievable temperature is caused by the exponential-like

temperature dependence of the equilibrium vapor pressure, as predicted by the Clausius-Clapeyron relation (10.21).

12.5) INITIAL TEMPERATURE FOR DEMAGNETIZATION COOLING

The sum of the entropies of the magnetic moments and of the phonons must be constant during isentropic demagnetization. The entropy σ_m of N_m magnetic moments of magnitude m in a magnetic field B may be obtained from (2.41),

$$\sigma_m = \sigma_0 - U^2/2m^2B^2N_m = \sigma_0 - m^2B^2N_m/2\tau^2, \quad (2.41)$$

where the second equality follows from the relation $1/\tau = (\partial\sigma/\partial U)_B = -U/m^2B^2N_m$; see Problem 2.2. If we neglect the effective interaction field B_Δ between the paramagnetic ions, the change in magnetic entropy during complete demagnetization is

$$\Delta\sigma_m = + N_m(mB)^2/2\tau_i^2 = (N_m/2)(T_m/T_i)^2,$$

where $T_i = \tau_i/k_B$ is the initial temperature and $T_m = mB/k_B$. For $m = \mu_B$ and $B = 100$ kG, $T_m = 6.717$ K.

The phonon entropy σ_p for $T \ll \theta$ is most easily obtained from the low-temperature heat capacity expression (4.47a) for a Debye solid of N_p atoms,

$$C_V = (12\pi^4 N_p/5)(T/\theta)^3,$$

by utilizing the relation (3.17a), $C_V = \tau(\partial\sigma/\partial\tau)$. This leads to

$$\sigma_p = (4\pi^4 N_p/5)(T/\theta)^3.$$

The change in phonon entropy in cooling from T_i to the final temperature T_f is

$$\Delta\sigma_p = -(4\pi^4 N_p/5)(T_i^3 - T_f^3)/\theta^3.$$

Setting $\Delta\sigma_m + \Delta\sigma_p = 0$ and solving for T_f leads to

$$T_f = T_i [1 - (T_0/T_i)^5]^{1/3}, \quad (S1)$$

where

$$T_0 = [(5/8\pi^4)(N_m/N_p)T_m^2\theta^3]^{1/5}.$$

If we ignore any differences between N_m and N_p , use the above value of T_m and set $\theta = 100$ K, we find $T_0 = 12.37$ K.

As T_i approaches T_0 from above, significant cooling sets in very abruptly. For $T_i = 1.1 T_0$, we obtain $T_f \cong 0.72 T_i$, not significant by our definition. To obtain $T_f = 0.1 T_i$ we must have, from (S1), $(T_0/T_i)^5 = (1-0.1)^3 = 0.999$, or $T_i = 1.0002 T_0$. For $T_i = T_0$, (S1) predicts $T_f = 0$, but in this case (or for $T_i < T_0$) our neglect of the effective interaction field B_Δ is no longer valid. The amount of cooling obtainable is then limited by B_Δ , rather than by the negligible phonon contribution to the entropy.